Florida International University
Summer 2025 Capstone 1 Project
HMT - Hardware Monitoring Tool

Students: Santiago Defelice, Daniel Adriazola, Louis Flores, Jonathan Gonzalez, Justin Wissart Jahiem
Instructor: Seyedmasoud Sadjadi

CURRENT SYSTEM

Most users don't monitor their hardware until something breaks, and those who do juggle multiple disconnected tools—Task Manager for CPU/RAM, third-party apps for temperatures, separate utilities for disk health and GPU stats. Each has its own interface with no unified view or historical data. Existing solutions require technical knowledge to interpret metrics, manual configuration of alerts, and remembering to actually check them regularly. Users are left guessing what's normal, what's dangerous, and when to act. Your tool eliminates this complexity: simply download and install, and monitoring begins automatically with no configuration required.

PROBLEM

Users risk data loss and system failures due to undetected hardware issues. Existing monitoring solutions are either too complex for average users or too simplistic to catch problems early, and they don't provide actionable insights tailored to individual system configurations.

REQUIREMENTS

Functional Requirements

- Any operating system has to be compatible with the HMT

Non-Functional Requirements

- UI must be responsive and intuitive for users
- Monitoring should be accurate and should not return false values

SYSTEM DESIGN

The tool is built using Python with a straightforward client-side architecture. It uses system libraries to collect hardware metrics (CPU, GPU, RAM, disk, temperatures) at regular intervals, processes this data locally, and presents it through a user interface. The design prioritizes simplicity and low resource overhead to avoid impacting system performance while monitoring

VERIFICATION

- Check for accurate data points such as temperature, clock speed, storage utilization percentage.
- Ensure that the tool works on all operating systems by installing on Mac, Windows, and Linux.

IMPLEMENTATION

The tool is built entirely in Python using a client-side architecture. It leverages system libraries like psutil for CPU, RAM, and disk metrics, and GPUtil for GPU monitoring to collect hardware data at regular intervals. The monitoring agent runs as a lightweight background process on the user's machine, gathering telemetry without significantly impacting system performance. All data processing and storage occur locally, ensuring user privacy. The collected metrics are displayed through a simple user interface that shows real-time status and historical trends. The implementation prioritizes simplicity—requiring zero configuration from users—while maintaining low resource overhead. The single-language Python stack ensures easy maintenance, cross-platform compatibility, and straightforward deployment via a simple download-and-run installer.

SUMMARY

This tool provides real-time visibility into your computer's hardware health and performance. It tracks critical metrics like CPU usage, temperatures, RAM consumption, disk space, and GPU performance—all in one place. By continuously monitoring these components, it helps users detect potential hardware failures early, identify performance bottlenecks, and understand how their system is actually being used. The tool eliminates the need for technical expertise or multiple applications, making hardware monitoring accessible to everyone. Simply install it and gain immediate insight into your system's condition, preventing unexpected crashes, data loss, and performance degradation before they happen.